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Practical 14

File Upload and Download Automation

Technology: Selenium WebDriver, Java, JUnit, Maven and Java Robot Class

Environment: GitHub Codespaces — Ubuntu Linux

1. Aim

To automate file upload and download operations using Selenium WebDriver, verify the uploaded filename and downloaded file content, and demonstrate the Java Robot Class.

2. Open the Practical Repository

1. Sign in to GitHub using your student account.
2. Open the repository –
https://github.com/Conceicaodsouza/STQA-Practical-14-FileUploadDownload
3. Open the repository in GitHub Codespaces.
4. Wait for the Codespace to load.
5. Open the terminal.

3. Verify the Working Directory

Execute: **pwd**

Expected output:

/workspaces/STQA-Practical-14-FileUploadDownload

If you are not in this directory, navigate to the repository folder before continuing.

4. Check Java Installation

Execute : **java -version**
Java 21 or later is recommended for this practical. The tested environment uses Microsoft OpenJDK 25.
Check the Java compiler : **javac -version**

Both commands should display the installed Java version.

If Java is missing

On Ubuntu, install Java 21:

sudo apt-get update
sudo apt-get install -y openjdk-21-jdk

Verify the installation:

java -version
javac -version

5. Check Maven Installation

Execute : **mvn -version**

The output should display the Maven version, Java version, and operating system.

If Maven is missing

Execute:
sudo apt-get update
sudo apt-get install -y maven

Verify : **mvn -version**

6. Check Google Chrome and ChromeDriver

Selenium requires a browser and a compatible driver. This practical uses Google Chrome for Testing and ChromeDriver.

Check whether Chrome is available in the system path:

which google-chrome

Check Chromium : **which chromium**

which chromium-browser

Check ChromeDriver : **which chromedriver**

These commands may return no output in Codespaces. This does not necessarily mean that Chrome is unavailable.

Check the Selenium browser cache:

ls -d ~/.cache/selenium/chrome/linux64/* 2>/dev/null

Check the ChromeDriver cache:

ls -d ~/.cache/selenium/chromedriver/linux64/* 2>/dev/null

If the cache directories do not exist yet, Selenium Manager can download the required browser and driver automatically when the tests are run.

Important: Do not manually install an unrelated ChromeDriver version. Selenium Manager handles the browser and driver setup for this practical.

7. Check the Ubuntu Version

Execute:
cat /etc/os-release | head

The tested environment uses Ubuntu 24.04 LTS.

8. Install Required Chrome Dependencies

Chrome requires certain Linux libraries to run in Codespaces.

Check whether the libraries are available:

ldconfig -p | grep -E 'libatk-1.0.so.0|libatk-bridge-2.0.so.0|libcups.so.2|libxkbcommon.so.0|libasound.so.2|libgbm.so.1|libXcomposite.so.1|libXdamage.so.1|libXfixes.so.3|libXrandr.so.2|libatspi.so.0'

If these dependencies are missing, install them using:

sudo apt-get update && sudo apt-get install -y libatk1.0-0 libatk-bridge2.0-0 libcups2 libxkbcommon0 libasound2t64 libgbm1 libxcomposite1 libxdamage1 libxfixes3 libxrandr2 libatspi2.0-0

Verify again:

ldconfig -p | grep -E 'libatk-1.0.so.0|libatk-bridge-2.0.so.0|libcups.so.2|libxkbcommon.so.0|libasound.so.2|libgbm.so.1|libXcomposite.so.1|libXdamage.so.1|libXfixes.so.3|libXrandr.so.2|libatspi.so.0'

The output should list the available libraries.

9. Verify Python 3

Python is used to start the local web server required by this practical.

Execute : **python3 --version**

If Python 3 is missing, install it:

sudo apt-get update
sudo apt-get install -y python3

Verify : **python3 --version**

10. Check the Practical Files

Execute : **ls**

Verify the Maven project file : **ls pom.xml**

Check the Java test file:

ls src/test/java/practical/FileUploadDownloadTest.java

Check the HTML page:

ls src/test/resources/file-upload-download.html

Check the sample upload file:

ls src/test/resources/upload-files/sample-upload.txt

Check the sample download file:

ls src/test/resources/download-files/sample-download.txt

All five files should be present.

11. Start the Local Web Server

Important: Start the web server before running mvn test. The test opens the webpage at http://127.0.0.1:8000/file-upload-download.html.

In the first terminal, execute:

python3 -m http.server 8000 --directory src/test/resources

Expected message:

Serving HTTP on 0.0.0.0 port 8000 ...

Keep this terminal open and running.

If Python reports that port 8000 is already in use

Check whether a server is already running:

ps -ef | grep '[h]ttp.server 8000'

If the existing server is serving this practical's src/test/resources directory, keep using it. Do not start another server on the same port.

12. Verify the Web Server

Open a **second terminal** in the same Codespace. Keep the first terminal running the server.

Execute : **curl -I http://127.0.0.1:8000/file-upload-download.html**

Expected result:

HTTP/1.0 200 OK

Verify the sample download file:

curl -I http://127.0.0.1:8000/download-files/sample-download.txt

Expected result:

HTTP/1.0 200 OK

If you receive Connection refused, return to the first terminal and check that the server is still running.

View the webpage in a browser

1. Open the **Ports** tab in Codespaces.
2. Forward port 8000 if it has not been forwarded already.
3. Open the forwarded port's URL.

The practical webpage should display the file upload form and download link.

13. Run the Selenium Tests

In the second terminal, make sure you are in the project directory : **pwd**

Then execute : **mvn test**

On the first run, Maven may download dependencies, while Selenium Manager may download Chrome and ChromeDriver. Wait until the command finishes.

Expected results

The output should report:

- **File upload validation:** PASS
- **File download verification:** PASS
- **Robot Class:** Headless environment detected and handled
- **Tests run:** 3
- **Failures:** 0
- **Errors:** 0
- **Build:** SUCCESS

The final output should contain:

Tests run: 3, Failures: 0, Errors: 0, Skipped: 0
BUILD SUCCESS

The Robot Class message in Codespaces explains that keyboard simulation requires a desktop environment. The test handles this limitation gracefully; actual Robot keyboard simulation is not performed in headless Codespaces.

A Selenium warning about an unavailable Chrome DevTools Protocol (CDP) implementation may appear because the browser version is newer than the CDP implementation bundled with Selenium 4.34.0. If the tests still finish with BUILD SUCCESS, this warning is non-fatal.

14. Verify the Downloaded File

After a successful test run, execute:

ls -l target/downloads/

Expected file:

sample-download.txt

Verify its contents:

cat target/downloads/sample-download.txt

Expected content:

STQA Practical 14 - Sample file for download verification.

The downloaded file should exist and contain the expected text.

15. Stop the Web Server

After the tests are complete, switch to the first terminal where the Python server is running.

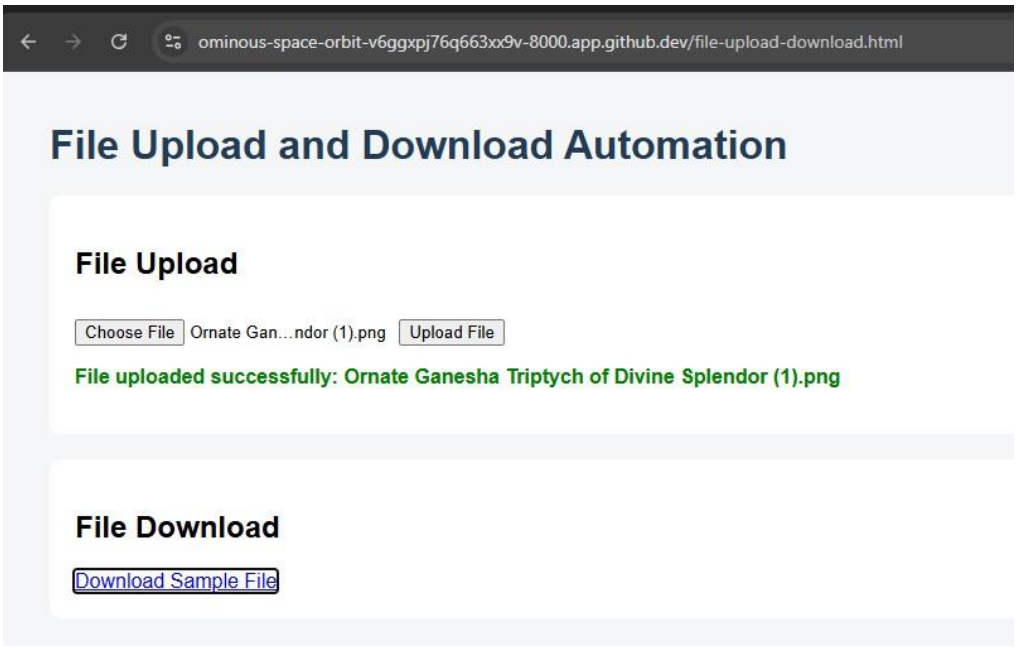
Press:

Ctrl + C

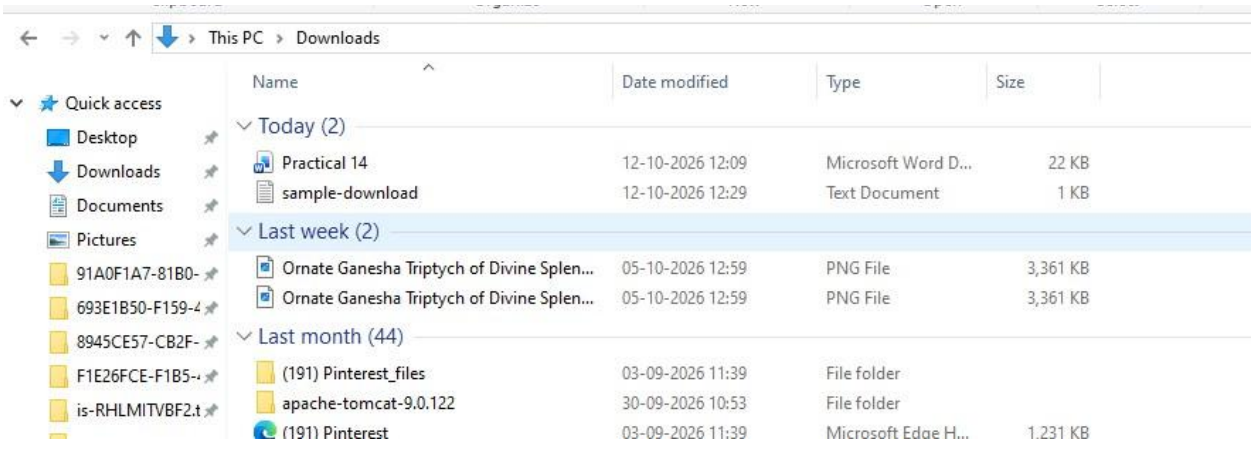
The web server will stop.

16. Result

File upload validation and file download verification were successfully automated using Selenium WebDriver and Java.



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```
[INFO] Running practical.FileUploadDownloadTest
Oct 12, 2026 7:05:47 AM org.openqa.selenium.devtools.CdpVersionFinder findNearestMatch
WARNING: Unable to find CDP implementation matching 155
Oct 12, 2026 7:05:47 AM org.openqa.selenium.chromium.ChromiumDriver lambda$new$4
WARNING: Unable to find version of CDP to use for 155.0.8059.39. You may need to include a dependency on a specific version o
f the CDP using something similar to `org.seleniumhq.selenium:selenium-devtools-v86:4.34.0` where the version ("v86") matches
 the version of the chromium-based browser you're using and the version number of the artifact is the same as Selenium's.
Starting file upload test...
PASS: File upload validation successful.
Oct 12, 2026 7:05:48 AM org.openqa.selenium.devtools.CdpVersionFinder findNearestMatch
WARNING: Unable to find CDP implementation matching 155
Oct 12, 2026 7:05:48 AM org.openqa.selenium.chromium.ChromiumDriver lambda$new$4
WARNING: Unable to find version of CDP to use for 155.0.8059.39. You may need to include a dependency on a specific version o
f the CDP using something similar to `org.seleniumhq.selenium:selenium-devtools-v86:4.34.0` where the version ("v86") matches
 the version of the chromium-based browser you're using and the version number of the artifact is the same as Selenium's.
Starting file download test...
PASS: Downloaded file exists and content is verified.
Oct 12, 2026 7:05:50 AM org.openqa.selenium.devtools.CdpVersionFinder findNearestMatch
WARNING: Unable to find CDP implementation matching 155
Oct 12, 2026 7:05:50 AM org.openqa.selenium.chromium.ChromiumDriver lambda$new$4
WARNING: Unable to find version of CDP to use for 155.0.8059.39. You may need to include a dependency on a specific version o
f the CDP using something similar to `org.seleniumhq.selenium:selenium-devtools-v86:4.34.0` where the version ("v86") matches
 the version of the chromium-based browser you're using and the version number of the artifact is the same as Selenium's.
Starting Robot Class test...
INFO: Robot keyboard simulation cannot run in headless Codespaces.
INFO: Robot demonstration requires a desktop environment.
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 5.041 s -- in practical.FileUploadDownloadTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  6.898 s
[INFO] Finished at: 2026-10-12T07:05:50Z
[INFO] ------------------------------------------------------------------------
@Dy1907 ➜/workspaces/STQA-Practical-14-FileUploadDownload (main) $
```

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Practical 15

TestNG Framework Implementation and Reporting

Tools: Selenium WebDriver, TestNG, Maven and Google Chrome

Step 1: Open the practical repository

1. Open the GitHub repository: https://github.com/Conceicaodsouza/STQA-Practical-15-TestNG-Framework
2. Click **Code → Codespaces → Create codespace on new**

Check that you are inside the practical folder:

pwd
ls

You should see pom.xml, README.md, src and testng.xml.

If you are not in the project folder, execute:

cd /workspaces/STQA-Practical-15-TestNG-Framework

Step 2: Check Java installation

Execute : **java -version**

Expected: Java version information, preferably Java 21 or later.

If Java is not found, do not proceed until Java is available. In the standard Codespaces environment used for this practical, Java 25 is already installed.

Step 3: Check Maven installation

Execute : **mvn -version**

Expected output includes:

- Apache Maven version
- Java version
- Operating system information

Step 4: Check Chrome and ChromeDriver

Run these commands one at a time:

which google-chrome
which chromium
which chromium-browser
which chromedriver

Important: These commands may return no output. This does not automatically mean the practical cannot run. Selenium Manager can download and manage Chrome and ChromeDriver when Maven runs the tests.

Check whether Selenium has already downloaded Chrome:

ls -la ~/.cache/selenium/

If the directory does not exist yet, continue to the next step.

Step 5: Install Chrome's required system dependencies

Chrome may fail to start in a new Codespace if required Linux libraries are missing.

Execute the following command:

sudo apt-get update && sudo apt-get install -y libatk1.0-0 libatk-bridge2.0-0 libcups2 libxkbcommon0 libasound2t64 libgbm1 libxcomposite1 libxdamage1 libxfixes3 libxrandr2 libatspi2.0-0

Wait until the installation finishes and the terminal prompt returns.

If the packages are already installed, that is fine.

If Chrome or ChromeDriver is not found: Do not try to install them manually. Proceed to Step 6; Selenium Manager normally handles the browser and driver download automatically during the first test run.

Step 6: Start the login webpage server

In **Terminal 1**, execute:

cd /workspaces/STQA-Practical-15-TestNG-Framework && python3 -m http.server 8000 --directory src/test/resources

Leave this terminal running. Do not press Ctrl+C.

The login page is available at port 8000.

Now, run the test in **Terminal 2:**

cd /workspaces/STQA-Practical-15-TestNG-Framework

mvn clean test

Step 7: Start the TestNG report server

Open a **third terminal**.

Execute:

cd /workspaces/STQA-Practical-15-TestNG-Framework && python3 -m http.server 8001 --directory target/surefire-reports

Leave this terminal running too.

The report server is available at port 8001. The report files will be generated when the tests run in the next step.

Step 8: Open the login page

1. Open the **Ports** tab in Codespaces.
2. If port 8000 is not listed, forward port 8000.
3. Click the globe/open-in-browser icon for port 8000.

The login page should display the title **STQA Student Login**.

Test credentials:

Field	Value
Username	student
Password	test123

The page should display a successful login message for the correct credentials and an error message for incorrect credentials.

The automated tests use a local HTML file, so the test execution does not depend on an external login website.

Step 9: Open the TestNG report

1. Return to the **Ports** tab.
2. If port 8001 is not listed, forward port 8001.
3. Click the globe/open-in-browser icon for port 8001.

4. Add /index.html to the end of the port's browser URL if necessary.

The report URL should have this general format:

https://<your-codespace-port-8001-url>/index.html

The report should display the test results. If it was opened before the tests finished, refresh the page after Maven completes.

Step 11: Verify the XML test results

Return to the terminal and execute:

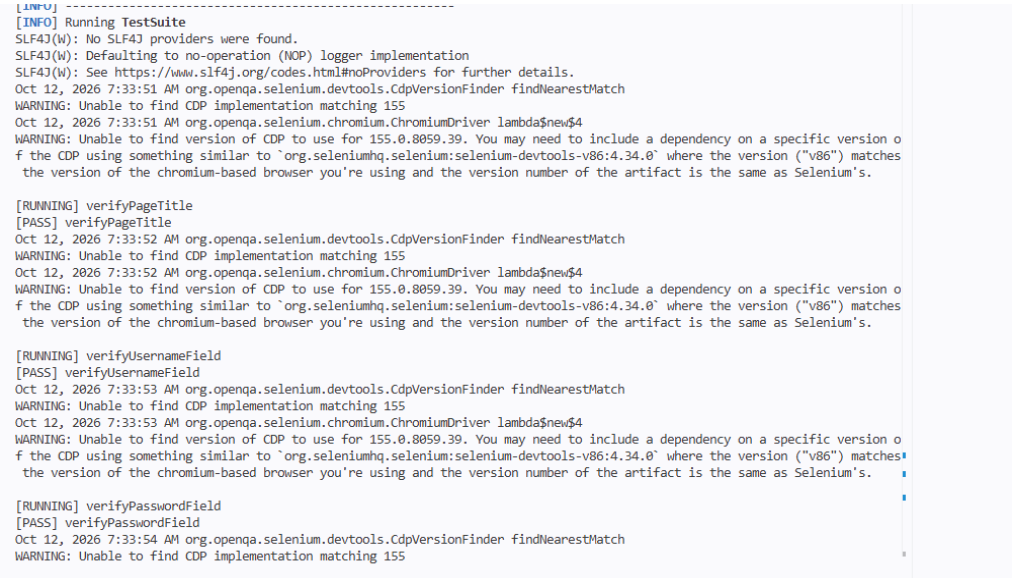
grep -E 'verifyPageTitle|verifyUsernameField|verifyPasswordField|verifySuccessfulLogin|verifyInvalidLogin' target/surefire-reports/testng-results.xml

To check the result counts, execute:

grep -E '<suite|<test |<test-method|total=|passed=|failed=|skipped=' target/surefire-reports/testng-results.xml

Confirm that all five test methods have status="PASS" and that there are no failures or errors.

Outputs:



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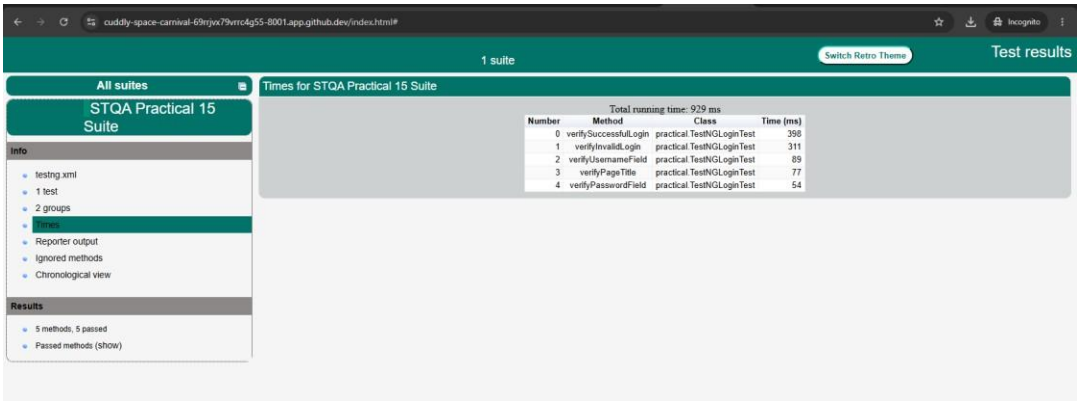
```
@Dy1907 ➜/workspaces/STQA-Practical-15-TestNG-Framework (main) $ mvn clean test
[PASS] verifyUsernameField
Oct 12, 2026 7:33:53 AM org.openqa.selenium.devtools.CdpVersionFinder findNearestMatch
WARNING: Unable to find CDP implementation matching 155
Oct 12, 2026 7:33:53 AM org.openqa.selenium.chromium.ChromiumDriver lambda$new$4
WARNING: Unable to find version of CDP to use for 155.0.8059.39. You may need to include a dependency on a specific version o
f the CDP using something similar to `org.seleniumhq.selenium:selenium-devtools-v86:4.34.0` where the version ("v86") matches
 the version of the chromium-based browser you're using and the version number of the artifact is the same as Selenium's.

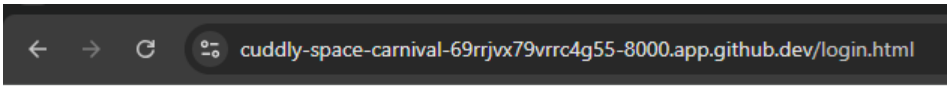
[RUNNING] verifyPasswordField
[PASS] verifyPasswordField
Oct 12, 2026 7:33:54 AM org.openqa.selenium.devtools.CdpVersionFinder findNearestMatch
WARNING: Unable to find CDP implementation matching 155
Oct 12, 2026 7:33:54 AM org.openqa.selenium.chromium.ChromiumDriver lambda$new$4
WARNING: Unable to find version of CDP to use for 155.0.8059.39. You may need to include a dependency on a specific version o
f the CDP using something similar to `org.seleniumhq.selenium:selenium-devtools-v86:4.34.0` where the version ("v86") matches
 the version of the chromium-based browser you're using and the version number of the artifact is the same as Selenium's.

[RUNNING] verifySuccessfulLogin
[PASS] verifySuccessfulLogin
Oct 12, 2026 7:33:56 AM org.openqa.selenium.devtools.CdpVersionFinder findNearestMatch
WARNING: Unable to find CDP implementation matching 155
Oct 12, 2026 7:33:56 AM org.openqa.selenium.chromium.ChromiumDriver lambda$new$4
WARNING: Unable to find version of CDP to use for 155.0.8059.39. You may need to include a dependency on a specific version o
f the CDP using something similar to `org.seleniumhq.selenium:selenium-devtools-v86:4.34.0` where the version ("v86") matches
 the version of the chromium-based browser you're using and the version number of the artifact is the same as Selenium's.

[RUNNING] verifyInvalidLogin
[PASS] verifyInvalidLogin
[INFO] Tests run: 5, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 7.272 s -- in TestSuite
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 5, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  10.560 s
[INFO] Finished at: 2026-10-12T07:33:56Z
[INFO] ------------------------------------------------------------------------
@Dy1907 ➜/workspaces/STQA-Practical-15-TestNG-Framework (main) $
```

python3
bash
python3





Student Login

Username: student

Password: •••••••

Login

Login successful!